



The LaTeX report

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1 Setup

1.1 Command history

```
ma5>import /Users/omattelaer/Documents/git_workspace/version3/PROC_sm_14/bin/internal/-
ufomodel
ma5>import /Users/omattelaer/Documents/git_workspace/version3/PROC_sm_14/Events/run_01/-
unweighted_events.lhe.gz as run_01
ma5>define vl = 12 14 16
ma5>define vl = -16 -14 -12
ma5>define invisible = ve vm vt vm ve vt vl vl
ma5>set main.graphic_render = matplotlib
ma5>plot THT 40 0 500 [logY]
ma5>plot MET 40 0 500 [logY]
ma5>plot SQRTS 40 0 500 [logY]
ma5>plot PT(l-[1]) 40 0 500 [logY]
ma5>plot ETA(l-[1]) 40 -10 10 [logY]
ma5>plot PT(l+[1]) 40 0 500 [logY]
ma5>plot ETA(l+[1]) 40 -10 10 [logY]
ma5>plot PT(p[1]) 40 0 500 [logY]
ma5>plot ETA(p[1]) 40 -10 10 [logY]
ma5>plot PT(p[2]) 40 0 500 [logY]
ma5>plot ETA(p[2]) 40 -10 10 [logY]
ma5>plot PT(p[3]) 40 0 500 [logY]
ma5>plot ETA(p[3]) 40 -10 10 [logY]
ma5>plot M(l+[1] p[1]) 40 0 500 [logY ]
ma5>plot M(l+[1] p[1] p[2]) 40 0 500 [logY ]
ma5>plot M(l+[1] p[1] p[2] p[3]) 40 0 500 [logY ]
ma5>plot M(l+[1] p[1] p[3]) 40 0 500 [logY ]
ma5>plot M(l+[1] p[2]) 40 0 500 [logY ]
ma5>plot M(l+[1] p[2] p[3]) 40 0 500 [logY ]
ma5>plot M(l+[1] p[3]) 40 0 500 [logY ]
ma5>plot M(l-[1] l+[1]) 40 0 500 [logY ]
ma5>plot M(l-[1] l+[1] p[1]) 40 0 500 [logY ]
ma5>plot M(l-[1] l+[1] p[1] p[2]) 40 0 500 [logY ]
ma5>plot M(l-[1] l+[1] p[1] p[2] p[3]) 40 0 500 [logY ]
ma5>plot M(l-[1] l+[1] p[1] p[3]) 40 0 500 [logY ]
ma5>plot M(l-[1] l+[1] p[2]) 40 0 500 [logY ]
ma5>plot M(l-[1] l+[1] p[2] p[3]) 40 0 500 [logY ]
ma5>plot M(l-[1] l+[1] p[3]) 40 0 500 [logY ]
ma5>plot M(l-[1] p[1]) 40 0 500 [logY ]
ma5>plot M(l-[1] p[1] p[2]) 40 0 500 [logY ]
ma5>plot M(l-[1] p[1] p[2] p[3]) 40 0 500 [logY ]
ma5>plot M(l-[1] p[1] p[3]) 40 0 500 [logY ]
ma5>plot M(l-[1] p[2]) 40 0 500 [logY ]
ma5>plot M(l-[1] p[2] p[3]) 40 0 500 [logY ]
ma5>plot M(l-[1] p[3]) 40 0 500 [logY ]
ma5>plot M(p[1] p[2]) 40 0 500 [logY ]
ma5>plot M(p[1] p[2] p[3]) 40 0 500 [logY ]
ma5>plot M(p[1] p[3]) 40 0 500 [logY ]
```

```

ma5>plot M(p[2] p[3]) 40 0 500 [logY ]
ma5>plot DELTAR(1+[1],p[1]) 40 0 10 [logY ]
ma5>plot DELTAR(1+[1],p[2]) 40 0 10 [logY ]
ma5>plot DELTAR(1+[1],p[3]) 40 0 10 [logY ]
ma5>plot DELTAR(1-[1],1+[1]) 40 0 10 [logY ]
ma5>plot DELTAR(1-[1],p[1]) 40 0 10 [logY ]
ma5>plot DELTAR(1-[1],p[2]) 40 0 10 [logY ]
ma5>plot DELTAR(1-[1],p[3]) 40 0 10 [logY ]
ma5>plot DELTAR(p[1],p[2]) 40 0 10 [logY ]
ma5>plot DELTAR(p[1],p[3]) 40 0 10 [logY ]
ma5>plot DELTAR(p[2],p[3]) 40 0 10 [logY ]
ma5>submit /Users/omattelaer/Documents/git_workspace/version3/PROC_sm_14/MA5_PARTON_ANALYSIS_analy

```

1.2 Configuration

- MadAnalysis version 1.9.60 (2026-03-17).
- Histograms given for an integrated luminosity of 10fb^{-1} .

2 Datasets

2.1 run_01

- Sample consisting of: [signal](#) events.
- Generated events: [10000](#) events.
- Normalization to the luminosity: [3002166](#)+/- [6712](#) events.
- **Ratio (event weight): 300 - warning: please generate more events (weight larger than 1)!**

Path to the event file	Nr. of events	Cross section (pb)	Negative wgts (%)
PROC_sm_14/Events/run_01/- unweighted_events.lhe.gz	10000	300 @ 0.22%	0.0

3 Histos and cuts

3.1 Histogram 1

* Plot: THT

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
run_01	3002166	1.0	157.848	106.6	0.0	1.66

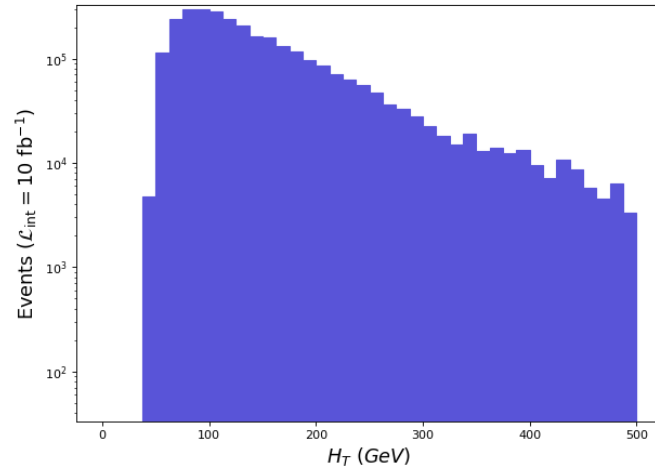


Figure 1.

3.2 Histogram 2

* Plot: MET

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
run_01	3002166	1.0	1.36849e-09	1.603e-09	0.0	0.0

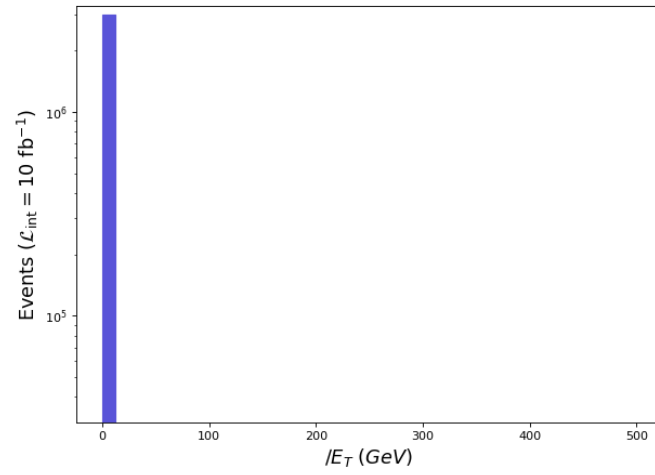


Figure 2.

3.3 Histogram 3

* Plot: SQRTS

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
run_01	3002166	1.0	550.34	377.6	0.0	41.58

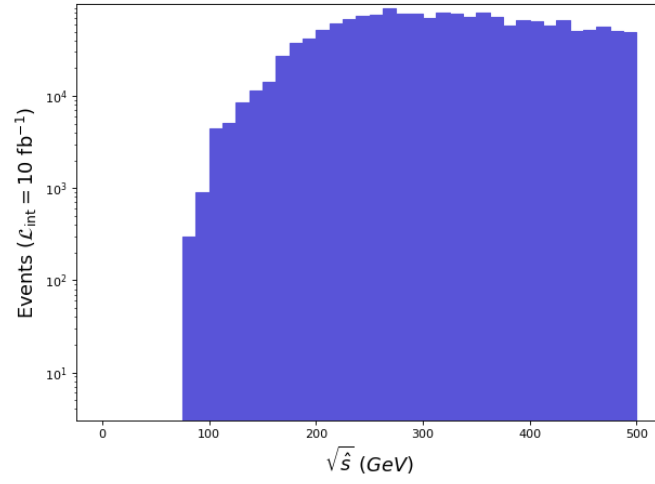


Figure 3.

3.4 Histogram 4

* Plot: PT (l-[1])

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
run_01	3002166	1.0	51.608	39.62	0.0	0.02

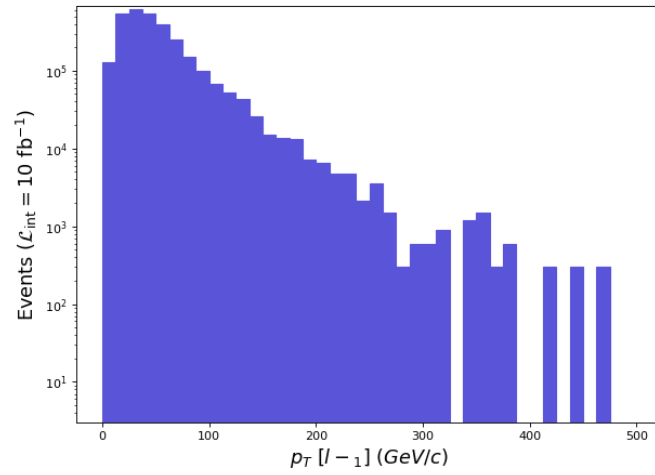


Figure 4.

3.5 Histogram 5

* Plot: $\text{ETA} \left(l-[1] \right)$

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
run_01	3002166	1.0	-0.00666294	1.339	0.0	0.0

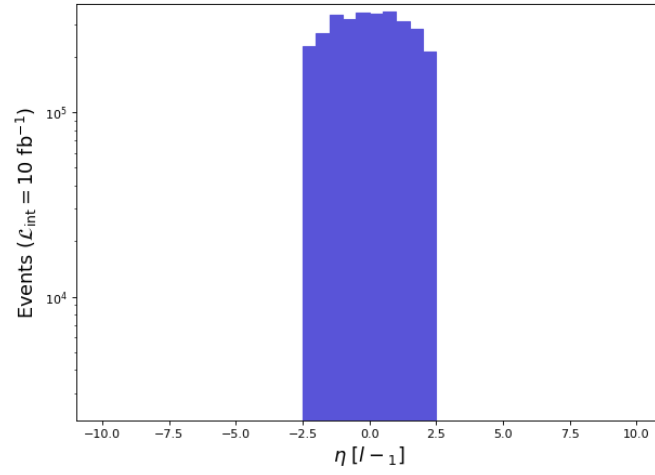


Figure 5.

3.6 Histogram 6

* Plot: PT (1+[1])

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
run_01	3002166	1.0	51.0089	38.66	0.0	0.0

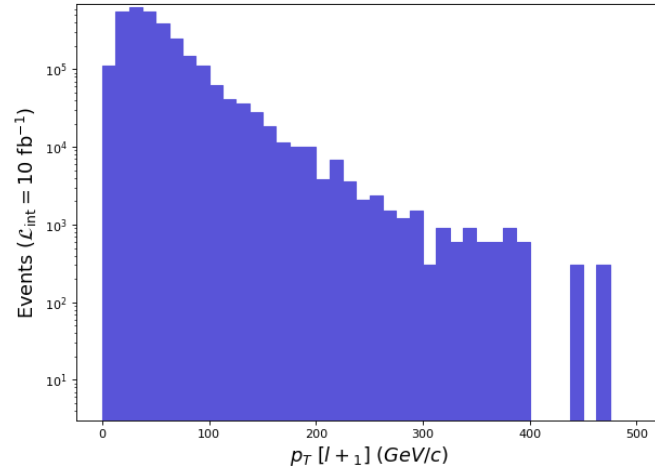


Figure 6.

3.7 Histogram 7

* Plot: $\text{ETA} (l+1)$

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
run_01	3002166	1.0	-0.00210033	1.316	0.0	0.0

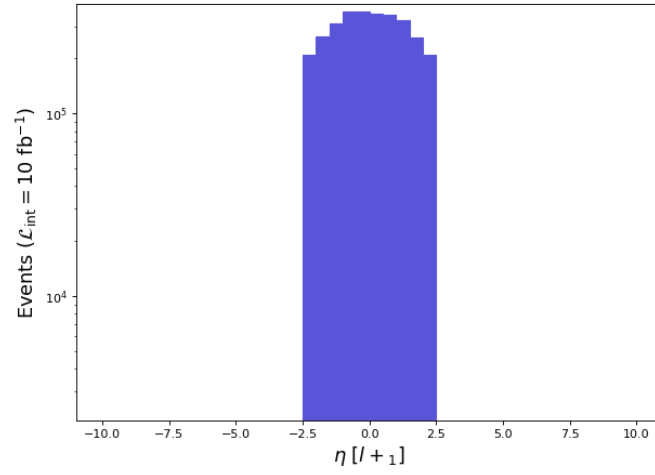


Figure 7.

3.8 Histogram 8

* Plot: PT (p[1])

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
run_01	3002166	1.0	84.2217	65.05	0.0	0.15

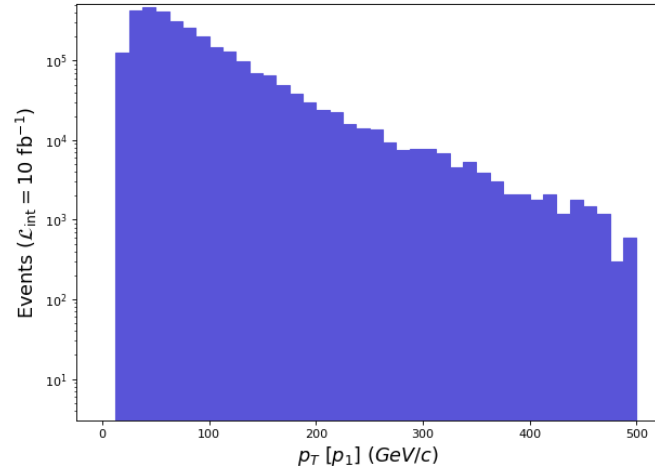


Figure 8.

3.9 Histogram 9

* Plot: $\text{ETA} (p[1])$

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
run_01	3002166	1.0	-0.0215702	1.711	0.0	0.0

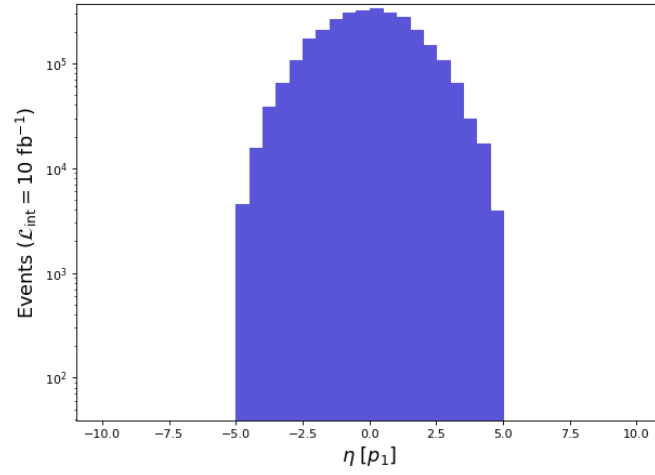


Figure 9.

3.10 Histogram 10

* Plot: PT (p[2])

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
run_01	3002166	1.0	47.8697	38.95	0.0	0.03

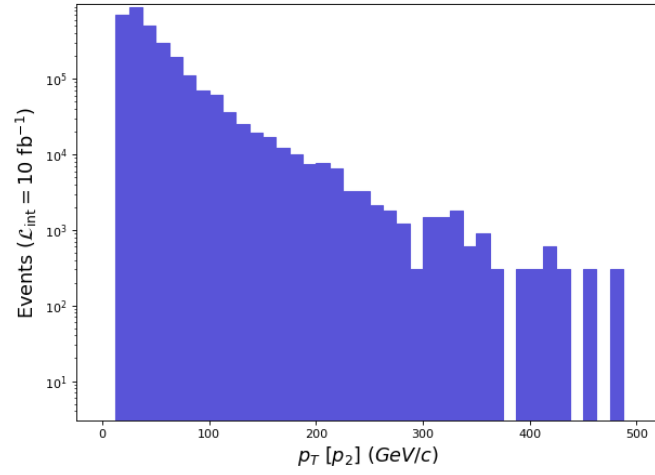


Figure 10.

3.11 Histogram 11

* Plot: ETA (p[2])

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
run_01	3002166	1.0	-0.0121327	1.902	0.0	0.0

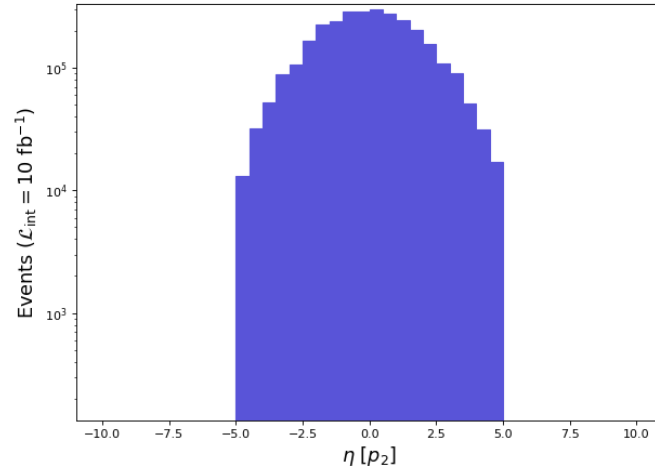


Figure 11.

3.12 Histogram 12

* Plot: PT (p[3])

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
run_01	3002166	1.0	25.7566	15.41	0.0	0.0

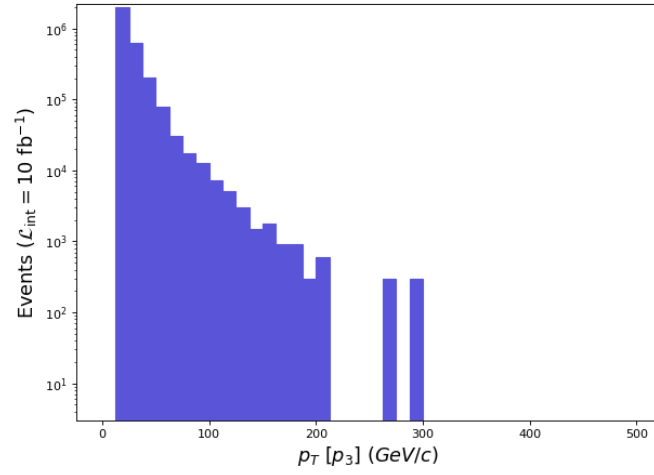


Figure 12.

3.13 Histogram 13

* Plot: ETA (p[3])

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
run_01	3002166	1.0	-0.0276682	2.085	0.0	0.0

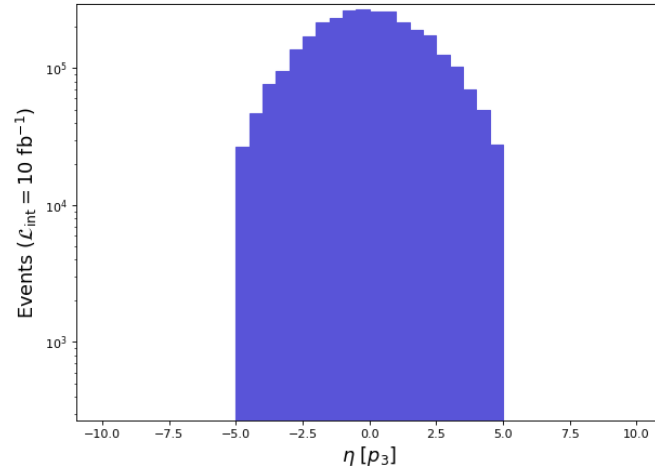


Figure 13.

3.14 Histogram 14

* Plot: $M(l + p_1)$

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
run_01	3002166	1.0	168.403	154.3	0.0	3.85

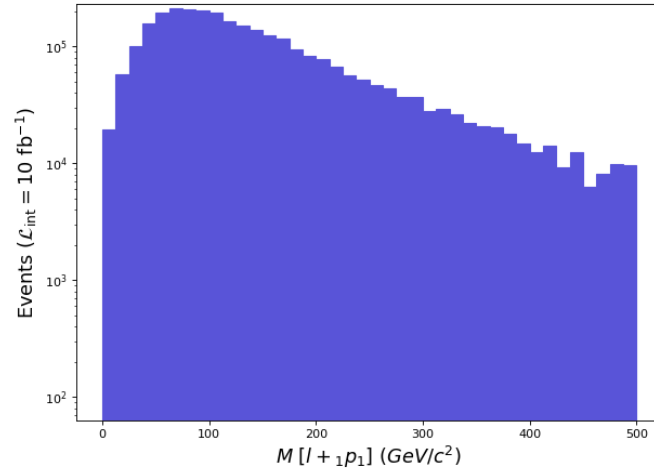


Figure 14.

3.15 Histogram 15

* Plot: $M(l + [1] p[1] p[2])$

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
run_01	3002166	1.0	328.045	271.0	0.0	16.54

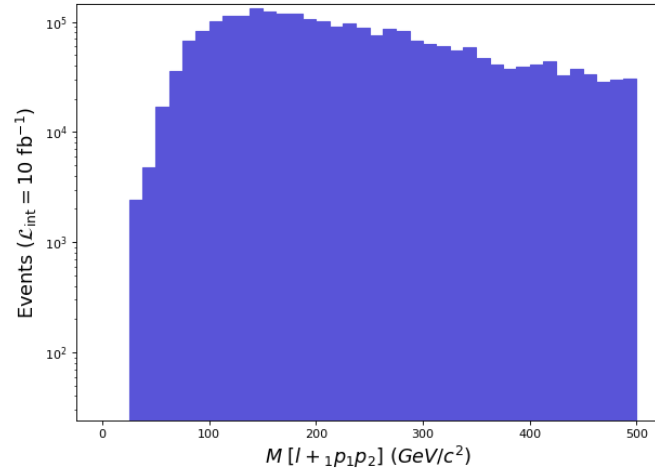


Figure 15.

3.16 Histogram 16

* Plot: $M(l + p_1 p_2 p_3)$

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
run_01	3002166	1.0	445.681	341.8	0.0	29.04

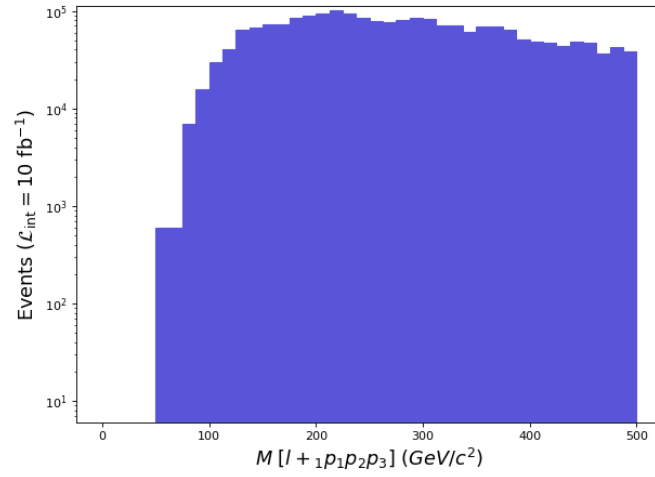


Figure 16.

3.17 Histogram 17

* Plot: $M(l + [1] p[1] p[3])$

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
run_01	3002166	1.0	289.564	244.6	0.0	12.7

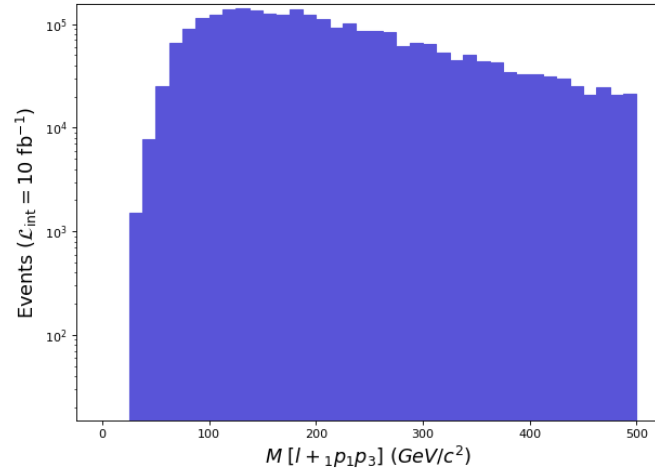


Figure 17.

3.18 Histogram 18

* Plot: $M(l + p_2)$

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
run_01	3002166	1.0	132.425	139.1	0.0	2.53

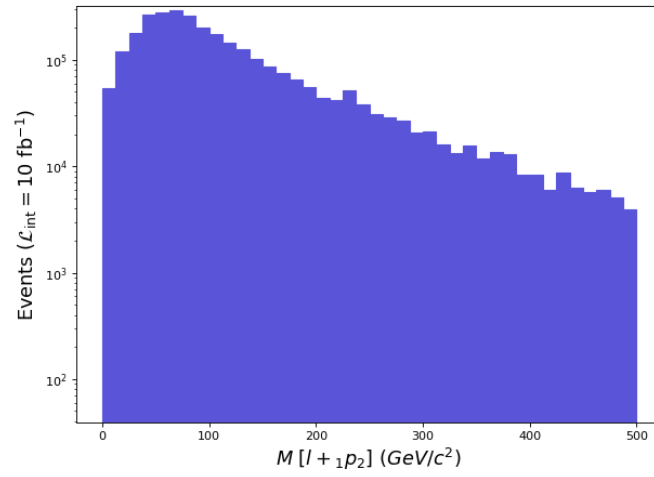


Figure 18.

3.19 Histogram 19

* Plot: $M (l + [1] p[2] p[3])$

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
run_01	3002166	1.0	246.862	222.1	0.0	9.61

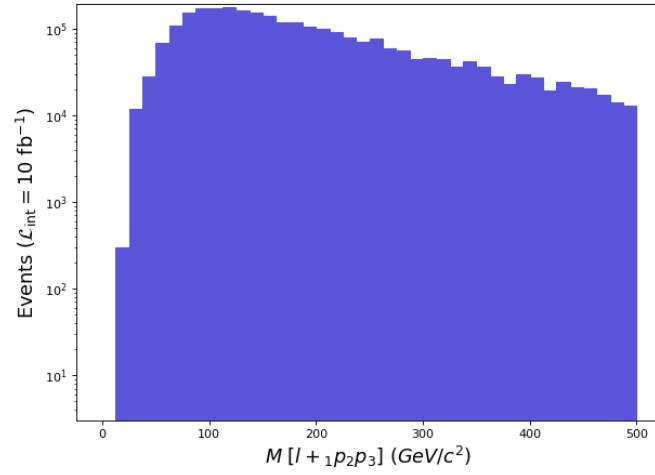


Figure 19.

3.20 Histogram 20

* Plot: $M(l + p_3)$

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
run_01	3002166	1.0	116.609	129.8	0.0	2.4

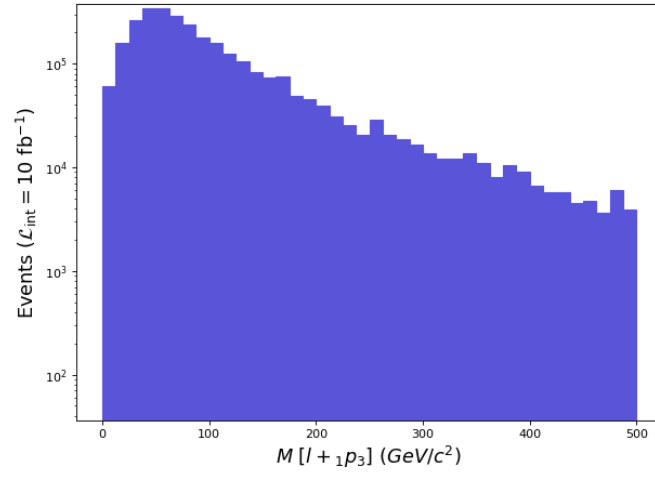


Figure 20.

3.21 Histogram 21

* Plot: $M(l_+ l_-)$

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
run_01	3002166	1.0	81.5113	30.5	0.0	0.03

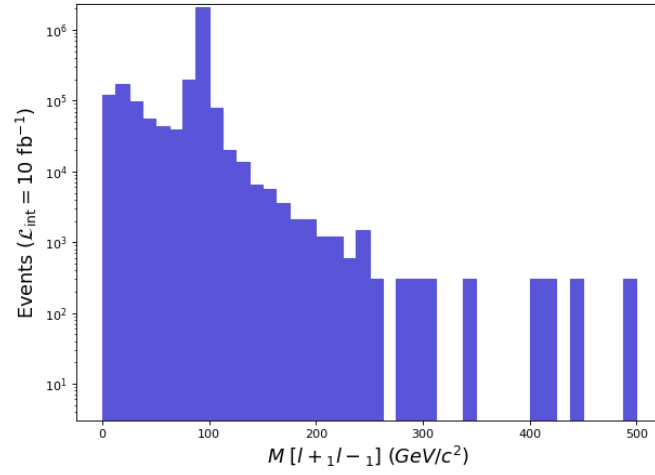


Figure 21.

3.22 Histogram 22

* Plot: $M(l^+l^-p_1)$

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
run_01	3002166	1.0	272.294	198.0	0.0	9.17

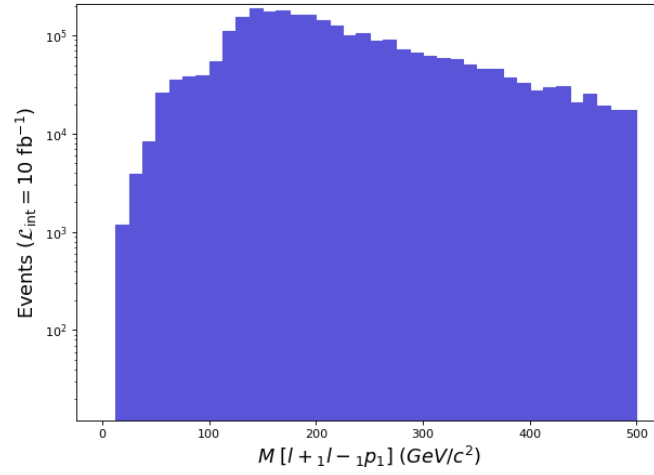


Figure 22.

3.23 Histogram 23

* Plot: $M(l^+l^-p_1p_2)$

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
run_01	3002166	1.0	430.849	308.8	0.0	26.93

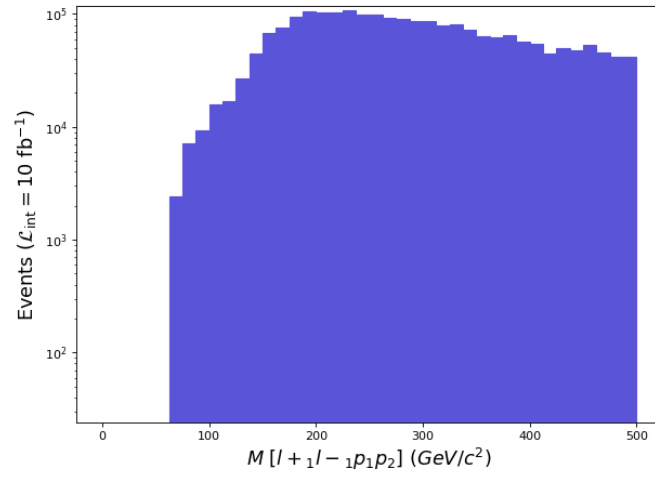


Figure 23.

3.24 Histogram 24

* Plot: $M(l+l-1 p_1 p_2 p_3)$

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
run_01	3002166	1.0	550.34	377.6	0.0	41.58

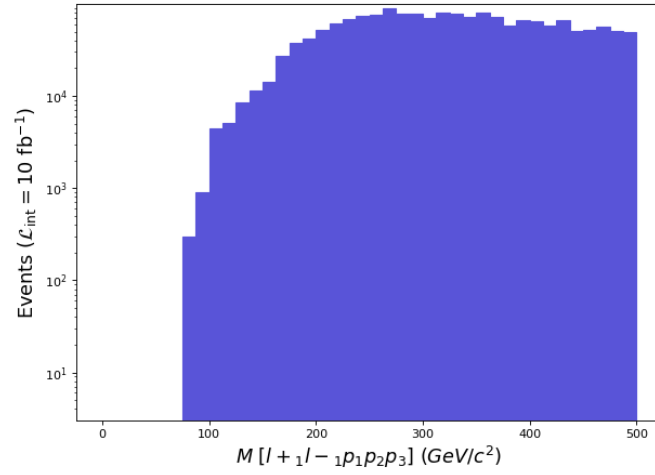


Figure 24.

3.25 Histogram 25

* Plot: $M(l^+l^-p_1p_3)$

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
run_01	3002166	1.0	393.489	283.0	0.0	21.91

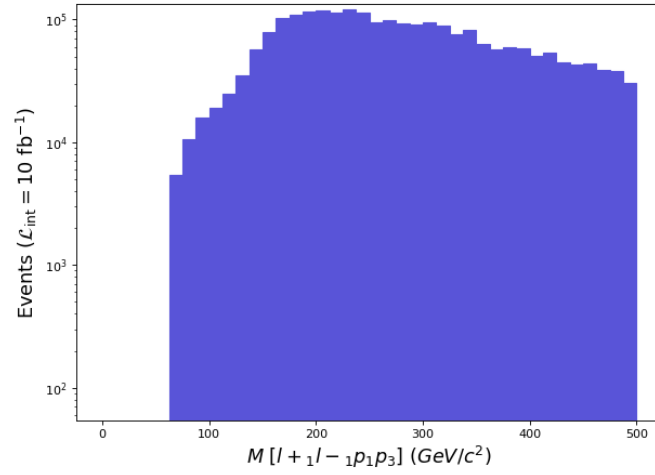


Figure 25.

3.26 Histogram 26

* Plot: $M(l^+l^-p_2)$

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
run_01	3002166	1.0	223.147	182.9	0.0	6.36

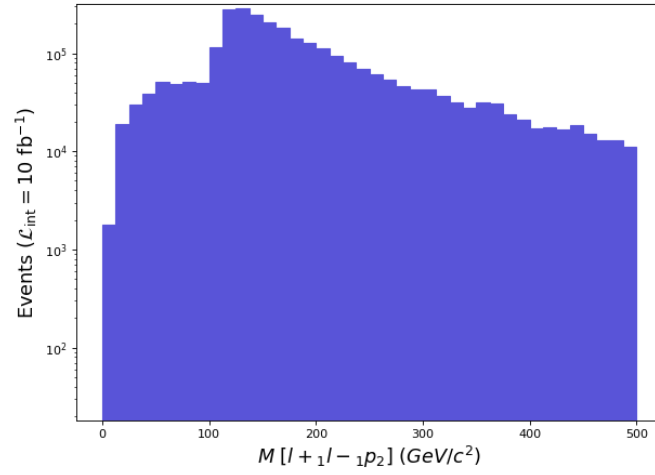


Figure 26.

3.27 Histogram 27

* Plot: $M (l^+l^-l^+l^-p_2p_3)$

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
run_01	3002166	1.0	343.233	262.4	0.0	17.17

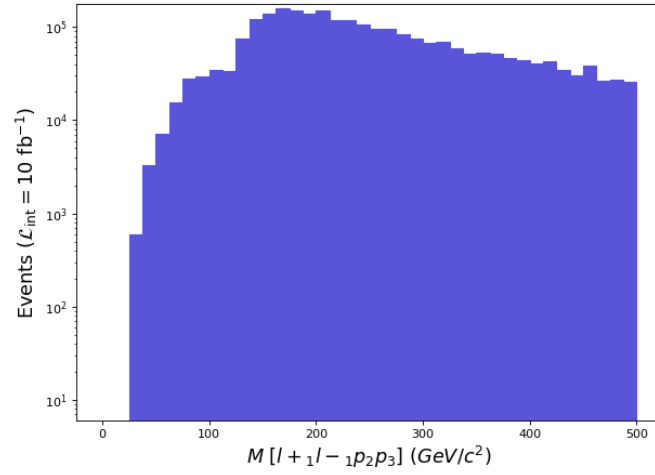


Figure 27.

3.28 Histogram 28

* Plot: $M(l^+l^-p_3)$

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
run_01	3002166	1.0	200.724	166.7	0.0	5.42

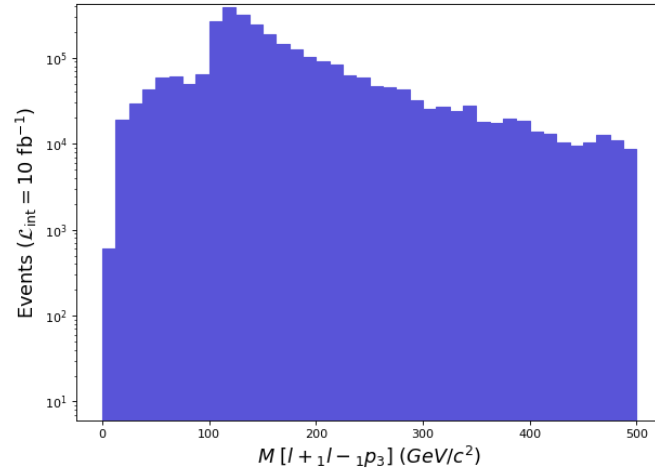


Figure 28.

3.29 Histogram 29

* Plot: $M(l^- p_1)$

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
run_01	3002166	1.0	171.187	155.9	0.0	3.86

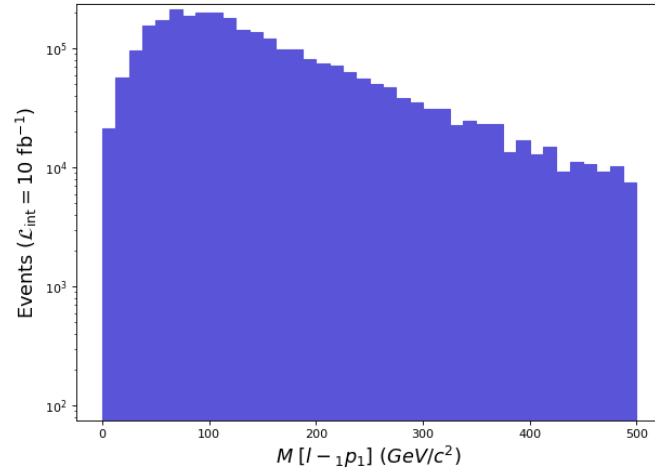


Figure 29.

3.30 Histogram 30

* Plot: $M (l^- l^+ p_1 p_2)$

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
run_01	3002166	1.0	332.179	272.0	0.0	17.1

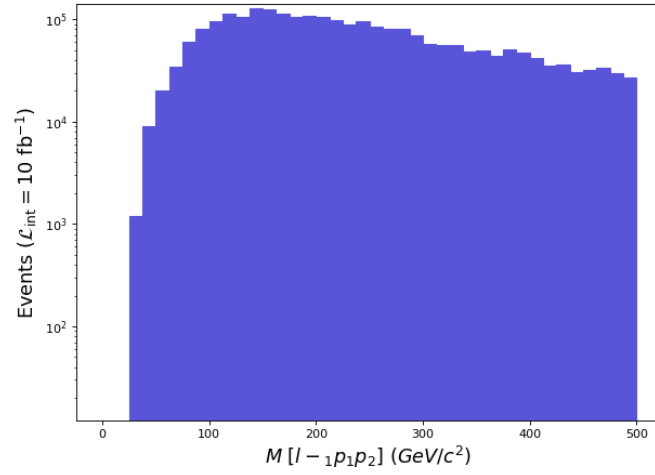


Figure 30.

3.31 Histogram 31

* Plot: $M(l^+l^-p_1p_2p_3)$

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
run_01	3002166	1.0	449.492	341.1	0.0	29.98

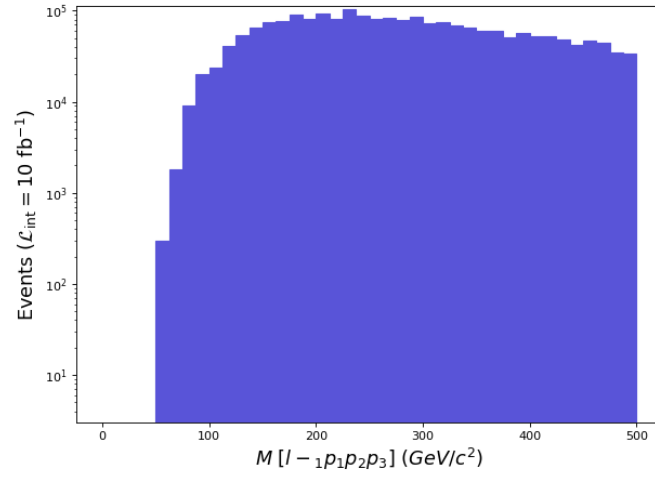


Figure 31.

3.32 Histogram 32

* Plot: $M (l^- l^+ p_1 p_3)$

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
run_01	3002166	1.0	292.436	243.3	0.0	12.76

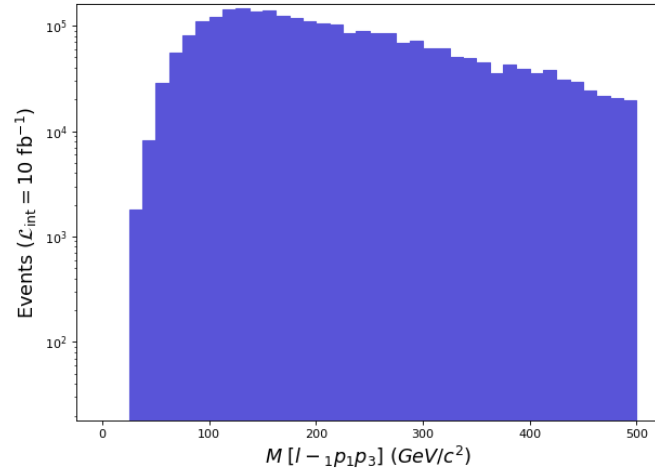


Figure 32.

3.33 Histogram 33

* Plot: $M(l_1 p_2)$

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
run_01	3002166	1.0	135.746	142.6	0.0	3.0

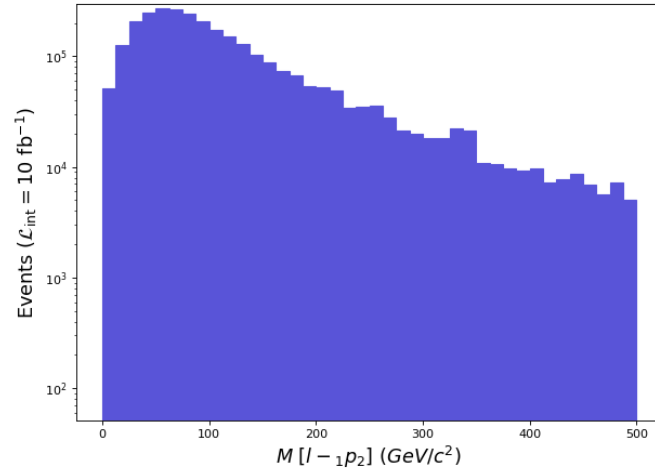


Figure 33.

3.34 Histogram 34

* Plot: $M (l^- l^+ p_2 p_3)$

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
run_01	3002165	1.0	250.156	221.7	0.0	9.98

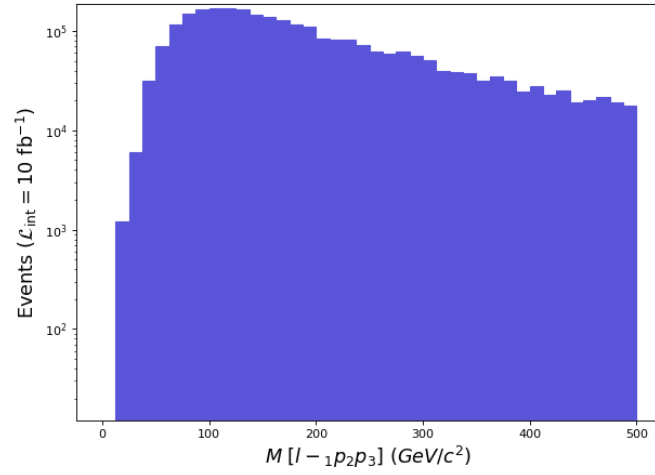


Figure 34.

3.35 Histogram 35

* Plot: $M(l^- p_3)$

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
run_01	3002166	1.0	117.602	127.4	0.0	2.25

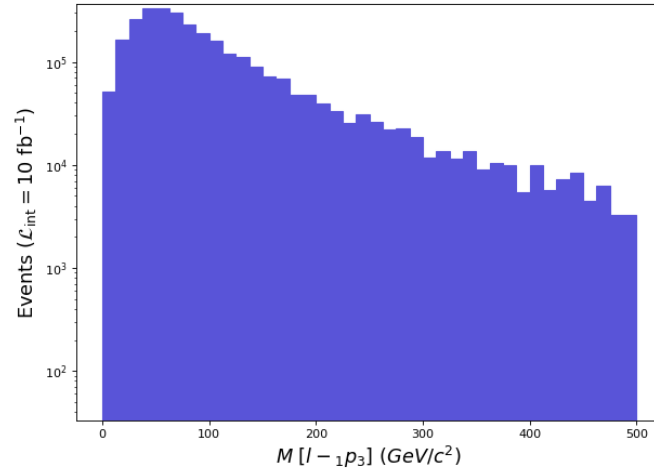


Figure 35.

3.36 Histogram 36

* Plot: $M(p_1 p_2)$

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
run_01	3002166	1.0	207.506	221.2	0.0	7.56

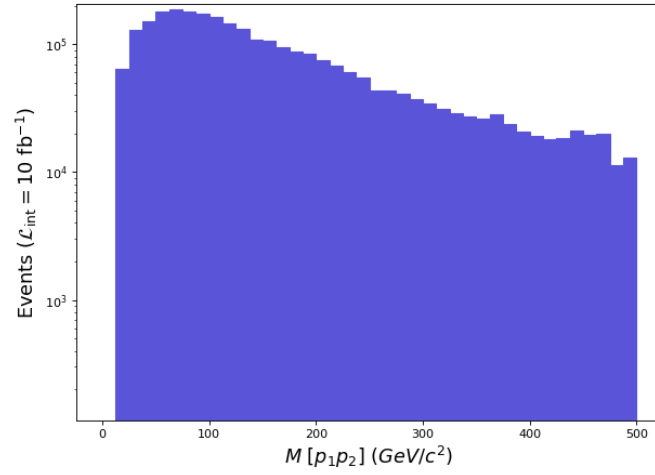


Figure 36.

3.37 Histogram 37

* Plot: $M (p[1] p[2] p[3])$

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
run_01	3002166	1.0	329.015	296.1	0.0	17.35

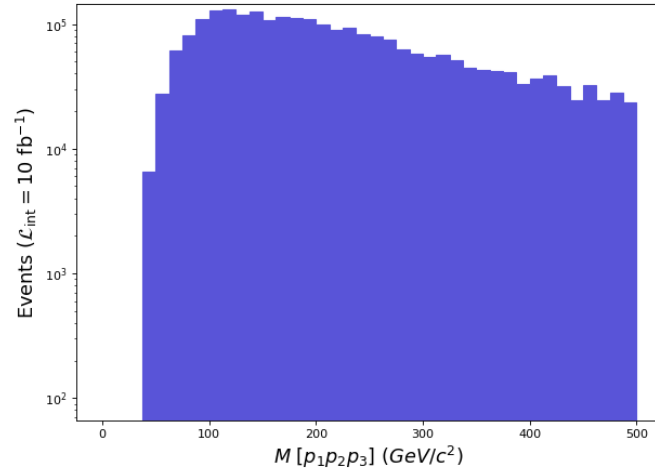


Figure 37.

3.38 Histogram 38

* Plot: $M(p_1 p_3)$

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
run_01	3002166	1.0	161.751	186.8	0.0	4.66

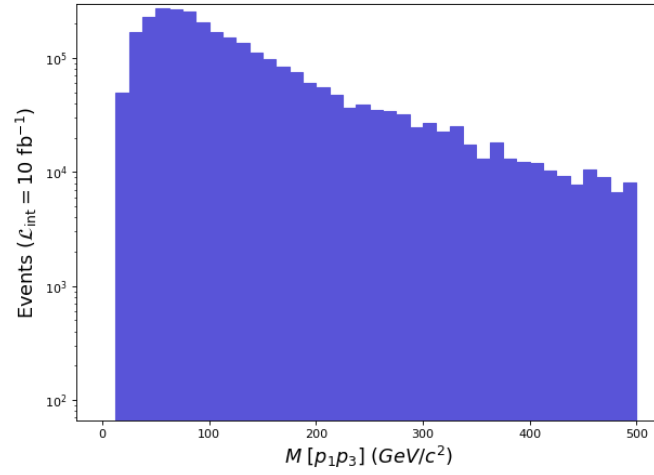


Figure 38.

3.39 Histogram 39

* Plot: $M(p_2 p_3)$

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
run_01	3002166	1.0	129.664	161.5	0.0	3.29

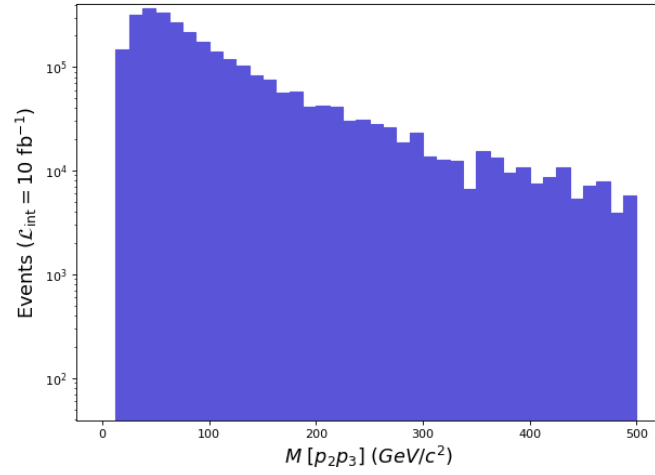


Figure 39.

3.40 Histogram 40

* Plot: DELTAR (l+[1] , p[1])

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
run_01	3002166	1.0	2.7177	1.016	0.0	0.0

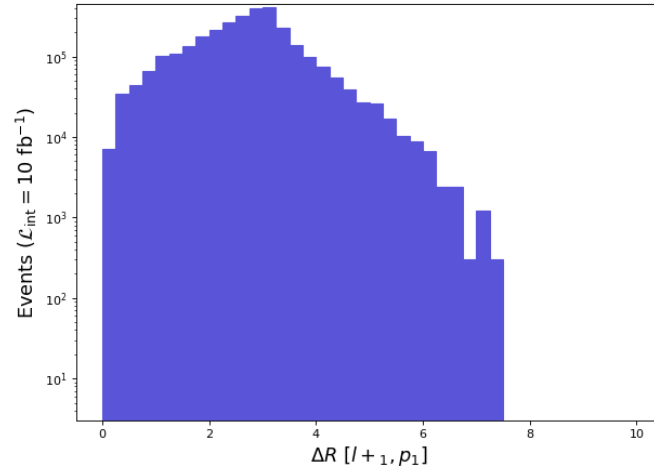


Figure 40.

3.41 Histogram 41

* Plot: DELTAR (l+[1] , p[2])

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
run_01	3002166	1.0	2.55415	1.198	0.0	0.0

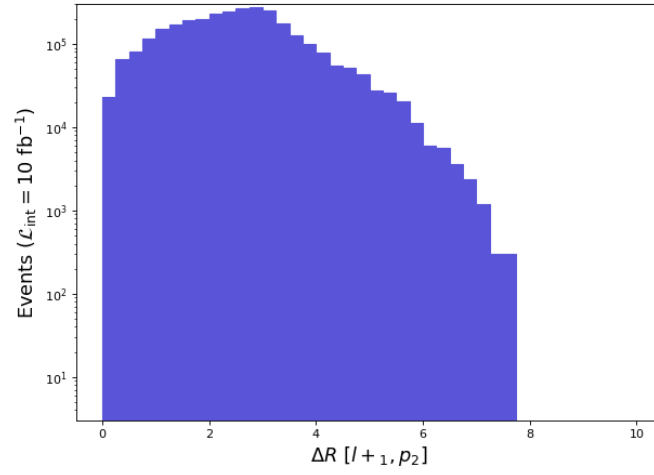


Figure 41.

3.42 Histogram 42

* Plot: DELTAR ($l+[1]$, $p[3]$)

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
run_01	3002166	1.0	2.72733	1.256	0.0	0.0

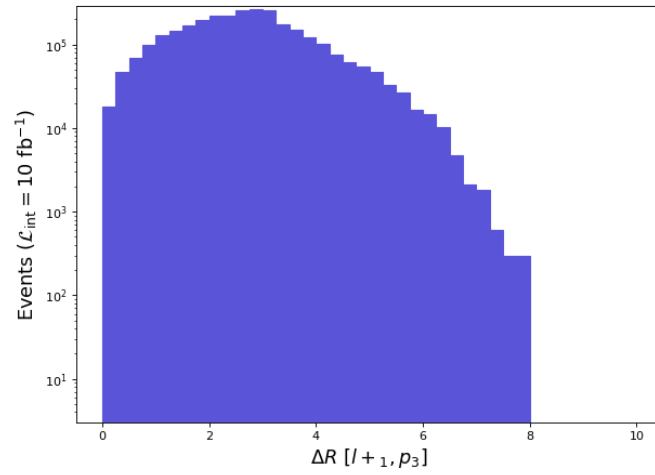


Figure 42.

3.43 Histogram 43

* Plot: DELTAR ($l-[1]$, $l+[1]$)

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
run_01	3002166	1.0	2.19402	0.8379	0.0	0.0

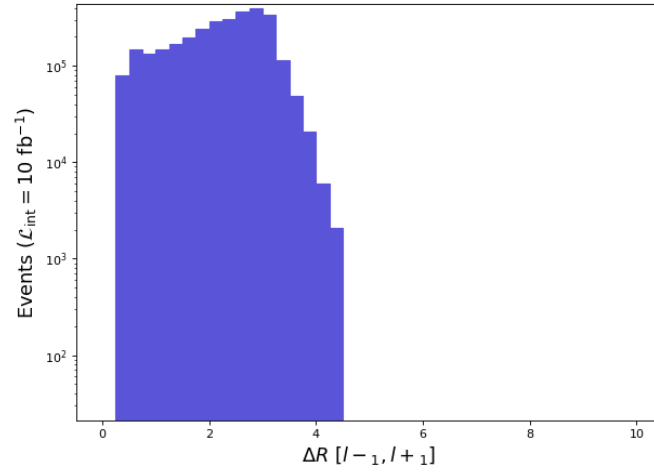


Figure 43.

3.44 Histogram 44

* Plot: DELTAR (l-[1] , p[1])

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
run_01	3002166	1.0	2.75042	1.021	0.0	0.0

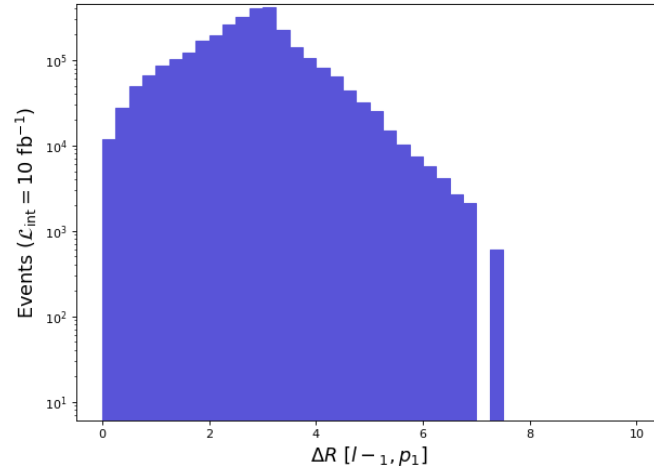


Figure 44.

3.45 Histogram 45

* Plot: DELTAR (l-[1] , p[2])

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
run_01	3002166	1.0	2.56444	1.225	0.0	0.0

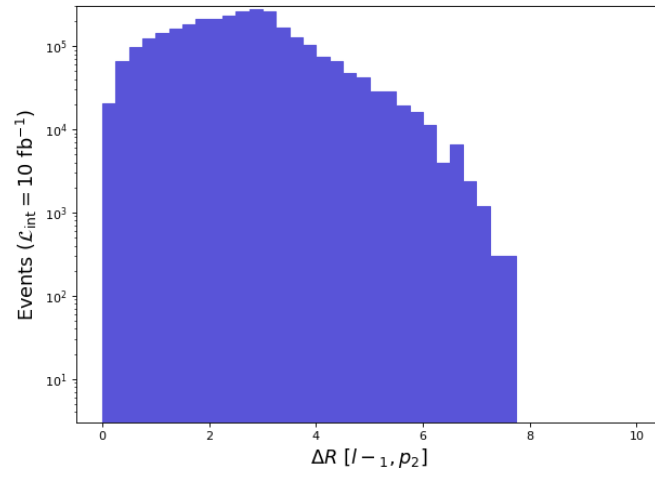


Figure 45.

3.46 Histogram 46

* Plot: DELTAR (l-[1] , p[3])

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
run_01	3002166	1.0	2.74493	1.264	0.0	0.0

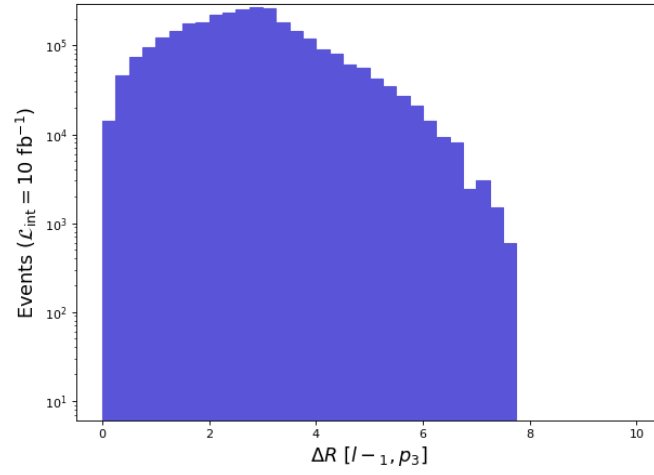


Figure 46.

3.47 Histogram 47

* Plot: DELTAR (p[1] , p[2])

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
run_01	3002166	1.0	2.95032	1.213	0.0	0.0

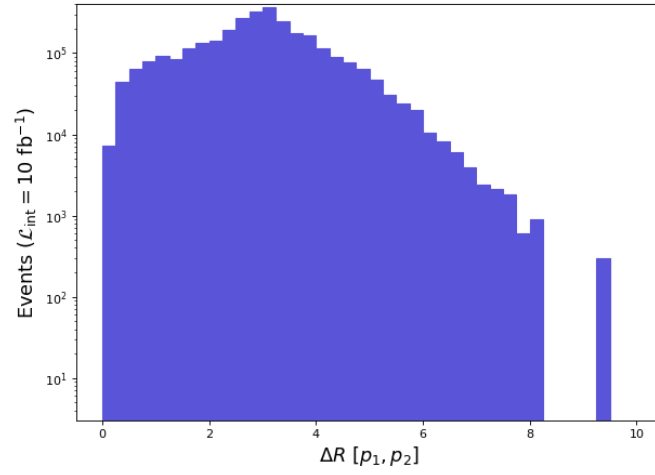


Figure 47.

3.48 Histogram 48

* Plot: DELTAR (p[1] , p[3])

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
run_01	3002166	1.0	2.88576	1.265	0.0	0.0

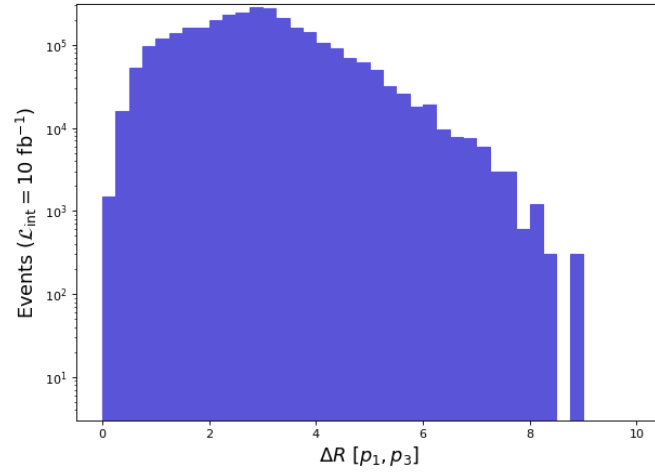


Figure 48.

3.49 Histogram 49

* Plot: DELTAR (p[2] , p[3])

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
run_01	3002166	1.0	2.85113	1.382	0.0	0.0

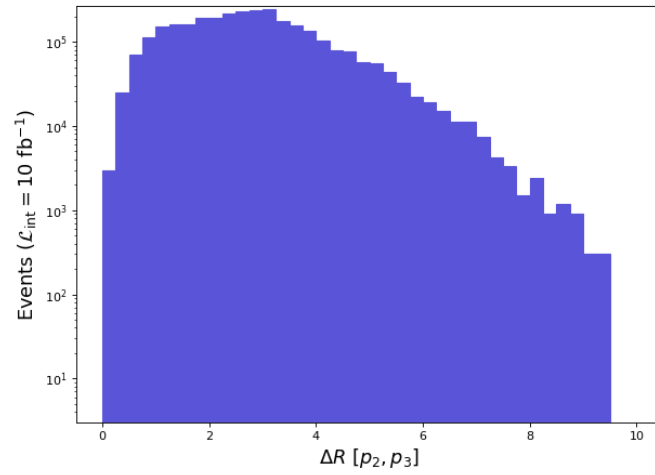


Figure 49.